Idejo — Spec Sheet v1

Oct 17, 2026 · @Mariano

1. Overview

Idejo version 1 is a self-hosted, open-source community server: one install hosts one community, organised like a phpBB forum, where every topic is a live chat. Administrators define categories and custom roles; members start topics; quick lines merge into “growing messages” so conversations read like a forum while feeling like a chat.

- **Operator:** anyone running their own server (association, project, club, company).
- **Readers:** members and, for public categories, anonymous visitors and search engines.
- **Basis:** this spec builds on the *Community Platform — Feature List* and the decisions in section 2.

Priority in requirement tables: **Must** = required for v1 release · **Should** = planned for v1, may slip to v1.x · **Could** = after v1. Requirement IDs (e.g. GM-03) are stable and used for tracking and tests.

2. Decisions

The twelve open questions from the feature list are settled as follows; each decision is reflected in the requirements below.

Question	Decision
Growing-message countdown	Appends do not restart the countdown. No size limit on a growing message. Default window 45 s, adjustable server-wide and per category.
Editing an open message	Allowed. If text already sent is changed (not just added to), a discreet “view earlier version” tag appears.
Line break key	Shift+Enter inserts a line break; Enter sends or appends; Ctrl+Enter closes.
When a topic stops being live	Optional auto-archive after 65 days of inactivity by default; period editable in server-wide settings.

Question	Decision
First message of a topic	Same editor as any message. Only differences: it has a title, it takes tags, and it is not live while being typed.
Replies inside a topic	Inline replies only. Anyone permitted can select any message, including someone else's, and create a new topic from it.
Communities per install	One community per server.
Licence	AGPL-3.0-only with a section 7(b) attribution notice. Copyright steward: Mariano Pictet until an organisation is formed, then transferred to it. Dual licensing kept possible.
Voice and video	Not in the product.
Federation	Not in v1; the architecture must allow it later.
Direct messages and encryption	No direct messages. Traffic is encrypted between client and server (TLS), not end-to-end.
Tech stack	Confirmed (section 18), provided the whole system can be built and maintained with Claude Code.
Product name and web address	Idejo, at idejo.org.
Server sizing	NF-04 and NF-05 accepted as targets.
Contributions and AI-generated code	Licence-grant CLA based on Apache templates; every change reviewed by a person, AI-assisted commits marked; lawyer review before any commercial licence (LI-05, LI-08).

3. Scope

Version 1 delivers text conversations in an admin-defined category structure, with roles, moderation, search and a public web presence.

In version 1

- Categories and sub-categories, topics, growing messages, inline replies, topic forking
- Custom roles and per-category permissions, managed by administrators
- Accounts, moderation, search, notifications (in-app, email, push)
- Public, search-engine-readable pages; RSS/Atom feeds

- Web admin panel, backups, upgrades, phpBB import
- Public API with real-time events; webhooks

Out of scope

- Direct or private messages between users
- Voice, video, screen sharing and stage events
- Several communities on one install
- End-to-end encryption
- Federation and bridges (architecture prepared, see section 15)
- Side threads inside topics

4. Information architecture

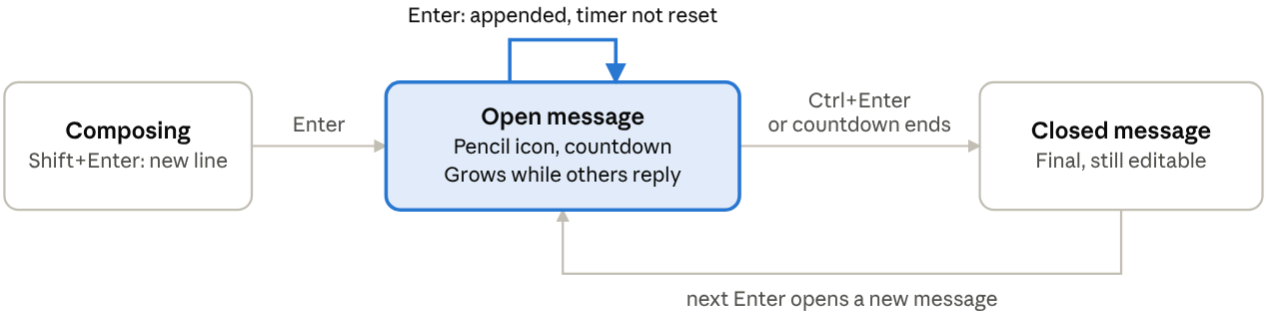
Four levels, each with a stable URL: server → category (nestable) → topic → message. Permissions are set on categories and flow down.

Entity	Created by	Key properties	URL pattern
Server	Installer	Name, branding, rules, registration mode, global settings	`/`
Category	Administrators (or delegated roles)	Name, description, order, parent category, visibility, permissions per role, growing-message window, auto-archive setting	`/c/{slug}`
Topic	Members with “start topics” right	Title, tags, state (open, locked, archived, solved), pinned flag, category	`/c/{slug}/t/{id}-{slug}`
Message	Members with “post” right	Author, body (Markdown), state (open or closed), commits, reply-to reference, reactions, attachments	`/c/{slug}/t/{id}-{slug}#m{id}`
Commit	System, at each Enter	Text fragment, timestamp, edited flag — the unit a growing message is built from	(internal)

5. Growing messages

A message stays open for a fixed window (45 s by default) after its first line is sent; every line sent in that window joins the same message. The window is not extended by appends, and a message has no size limit.

The window runs from the first line; appends never extend it



message states · compose, open, closed

Key	Action
Enter	Sends: starts a new message, or appends to the author's open message
Shift+Enter	Line break inside the composer, nothing sent
Ctrl+Enter (Cmd+Enter on macOS)	Sends any text in the composer and closes the open message
Touch screens	Send button = Enter; a close button beside it = Ctrl+Enter

ID	Requirement	Priority
GM-01	Enter with no open message posts a new message, open and visible to readers at once	Must
GM-02	Enter while the author has an open message in the topic appends the text as a new commit; readers see it appear in place	Must
GM-03	The window starts at the first commit; appends never restart or extend it	Must
GM-04	When the window ends the message closes; the next Enter starts a new message	Must

ID	Requirement	Priority
GM-05	Ctrl+Enter appends any composer text as the last commit and closes the message immediately	Must
GM-06	No size limit on a message; bodies longer than one screen render collapsed with a “show all” control	Must
GM-07	An open message shows a discreet pencil icon to all readers; the author also sees the time left	Must
GM-08	Messages are ordered by their first commit; replies by others appear below an open message, which keeps its place and keeps growing	Must
GM-09	An author has at most one open message per topic	Must
GM-10	The author can edit any commit, while open or after closing. Adding text at the end is not an edit	Must
GM-11	Changing text of an earlier commit marks the message as edited and shows a discreet “view earlier version” tag; it opens every previous version with its time	Must
GM-12	An open message closes early when its topic is locked or archived, or when the author’s session disconnects	Must
GM-13	Notifications, unread counts, search, feeds and webhooks treat a growing message as one item (see sections 10, 11, 15)	Must
GM-14	The first message of a topic is never live: it is composed privately and posted closed (see TP-02)	Must
GM-15	Window length set server-wide (default 45 s) with a per-category override; consolidation can be switched off per category	Must
GM-16	Moderators can close another user’s open message	Should

6. Topics

A topic is created like a forum thread and then runs as a live conversation of growing messages, with inline replies only.

ID	Requirement	Priority
TP-01	Members with the “start topics” right create a topic in a category with a title (required), tags and a first message	Must
TP-02	The first message uses the same editor as any message; nobody sees it while it is typed; it is posted closed together with the title and tags	Must
TP-03	Category view lists topics with title, starter, tags, message count, participants, last activity and unread marker; pinned topics on top	Must
TP-04	New messages and appends reach every viewer live, with automatic reconnect and catch-up of missed events	Must
TP-05	Full history kept; jump to the first message, the first unread or a date	Must
TP-06	Stable permalink to every topic and every message	Must
TP-07	Per-user read position per topic; “new since last visit” across categories; a growing message counts as one unread item	Must
TP-08	Inline replies: a reply references one message and shows a short excerpt linking to it. No side threads	Must
TP-09	Fork to new topic: a member with “start topics” in the target category selects any message, theirs or someone else’s, and starts a new topic from it; the new topic shows a linked excerpt, and the source message gets a discreet “continued in …” link	Must
TP-10	Mentions: @user, @role, @here (people in the topic now), @everyone; role and everyone mentions gated by permission	Must
TP-11	Composer: Markdown, code blocks with highlighting, spoilers, emoji picker, preview	Must
TP-12	Attachments and inline images with size and type limits per role; link previews fetched by the server	Must
TP-13	Follow topics or whole categories; bookmark single messages	Must

ID	Requirement	Priority
TP-14	Topic states open, locked, archived and solved; pin to the category or the whole server	Must
TP-15	Moderator tools: rename, move to another category, merge topics, split selected messages into a new topic	Must
TP-16	Presence: who is reading the topic right now	Should
TP-17	Reactions on messages; custom server emoji uploaded by admins	Should
TP-18	Mark the message that answered the topic as the solution, in Q&A categories	Should
TP-19	Tags and prefixes; required tags per category	Should
TP-20	Unsent composer text saved server-side and restored on any device	Should
TP-21	Polls inside a topic	Could

7. Roles, categories and permissions

Administrators manage custom roles (as in Discord) and categories (as in phpBB) from the admin panel; permissions are a matrix of roles × rights per category.

ID	Requirement	Priority
RP-01	Built-in roles always present: owner, administrator, moderator, member, guest	Must
RP-02	Role editor: create, rename, colour, icon, rank order and delete an unlimited number of custom roles	Must
RP-03	Category manager: create, nest, reorder by drag and drop, describe and delete categories; deleting asks whether to move or archive their topics	Must
RP-04	Permission matrix per category: view, read history, start topics, post, attach, react, mention roles, mention everyone, fork topics, pin, moderate	Must

ID	Requirement	Priority
RP-05	Sub-categories inherit their parent’s permissions; overrides set allow, deny or inherit per role or per user	Must
RP-06	Users hold several roles; their rights combine, an explicit deny on a category wins, the highest-ranked role sets the name colour	Must
RP-07	Moderator rights can be scoped to one category and its sub-categories	Must
RP-08	“View as”: an admin sees the server exactly as a chosen role or user sees it	Should
RP-09	“Manage roles” and “Manage categories” can be delegated; a delegate cannot grant rights or ranks above their own	Should
RP-10	Self-assigned roles chosen by members from an admin-defined menu	Should
RP-11	Automatic roles granted by trust level, account age or verification	Should
RP-12	Roles that expire on a set date	Could

8. Accounts and profiles

One account per person on the server; anonymous visitors can read public categories. There are no private messages between users.

ID	Requirement	Priority
AC-01	Sign-up with email and password, email verification	Must
AC-02	Registration modes: open, admin approval, invite-only, closed	Must
AC-03	Two-factor sign-in with authenticator codes and passkeys; recovery codes	Must
AC-04	Profile: display name, avatar, bio, pronouns, links, plus fields defined by admins	Must

ID	Requirement	Priority
AC-05	Visitors without an account read public categories and topics	Must
AC-06	Ignore a user: their messages are collapsed for you	Must
AC-07	Users export their data and delete their account themselves	Must
AC-08	List and revoke active sessions and devices	Must
AC-09	Single sign-on via OpenID Connect, SAML or LDAP	Should
AC-10	Trust levels unlocked by account age and activity, used against spam	Should
AC-11	Status: online, away, invisible	Should
AC-12	Profile badges	Could

9. Moderation

Moderators work from one queue, every action is logged, and filters apply to each commit of a growing message as it arrives.

ID	Requirement	Priority
MO-01	Report a message, topic or user with a reason; reports go to the moderation queue	Must
MO-02	Actions: hide, delete, edit, move, lock, close an open message; warn, time out, ban (temporary or permanent; by account, email or IP)	Must
MO-03	Searchable moderation log of every action, with moderator, target, reason and time	Must
MO-04	Approval queue for new topics and messages from new users or in chosen categories	Must
MO-05	Sign-up spam defence: captcha or proof-of-work, disposable-email blocklist, rate limits	Must
MO-06	Moderators see the full version history of any edited message	Must

ID	Requirement	Priority
MO-07	Automod rules: word, link and pattern filters checked on every commit, flood limits, an action per rule	Should
MO-08	Raid protection: join-rate limit and one-click lockdown	Should
MO-09	Private moderator notes on a user	Should
MO-10	One appeal per ban, reviewed in the queue	Could

10. Search, discovery and SEO

Public knowledge must be findable both inside the server and from search engines, which is what a chat-only tool cannot offer.

ID	Requirement	Priority
SE-01	Full-text search over topic titles and messages, filtered by author, date, category, tag and has-attachment	Must
SE-02	Results only show content the searcher may read	Must
SE-03	Messages are indexed as one document each; open messages are re-indexed as they grow	Must
SE-04	Public topics served as server-rendered HTML with clean URLs, sitemap, canonical links and discussion structured data; growing messages appear as single posts	Must
SE-05	RSS/Atom feeds per category, topic and tag, listing closed messages only	Must
SE-06	Topic lists: latest, most active, unanswered, per tag	Should
SE-07	Search adapter interface so a large server can plug in an external search engine	Should
SE-08	Related-topic and possible-duplicate suggestions when starting a topic	Could

11. Notifications

A growing message produces one notification, so a burst of quick lines never becomes a burst of alerts.

ID	Requirement	Priority
NO-01	In-app notification centre: mentions, replies, quotes, forks of your messages, activity in followed topics and categories	Must
NO-02	A growing message notifies once, when it opens; an append notifies only someone newly @mentioned in it	Must
NO-03	Levels per category and per topic: all, mentions only, none	Must
NO-04	Email: instant or daily/weekly digest, sent from closed messages only; one-click unsubscribe	Must
NO-05	Web push and mobile push without dependence on a single vendor relay	Should
NO-06	Quiet hours per user	Should

12. Accessibility and languages

The whole interface meets WCAG 2.2 level AA, including the live conversation view.

ID	Requirement	Priority
AX-01	Full keyboard operation, visible focus, landmarks and headings on every page	Must
AX-02	Screen readers announce a growing message when it opens and when it closes, not at every append	Must
AX-03	The pencil icon, the countdown and the “view earlier version” tag have text equivalents	Must
AX-04	Public pages readable without JavaScript	Must
AX-05	Every interface string translatable; right-to-left languages supported	Must

ID	Requirement	Priority
AX-06	Light, dark and high-contrast themes; text scaling; reduced motion	Should
AX-07	Per-user language, time zone and date format; alt-text prompt on image upload	Should

13. Administration and operations

After a one-command install, everything day-to-day is done in the web admin panel; the command line is only needed for recovery.

ID	Requirement	Priority
AD-01	One-command install with containers (app, database, reverse proxy with automatic certificates); first-run wizard creates the owner account	Must
AD-02	Web admin panel for settings, users, roles, categories, emoji, storage, email, appearance and legal pages	Must
AD-03	Auto-archive setting: off by default; when on, a topic with no new message for 65 days becomes archived (read-only, still searchable). The period is editable in server-wide settings; moderators can reopen a topic	Must
AD-04	Growing-message window: server default (45 s) and per-category override (GM-15)	Must
AD-05	Scheduled backups to local disk or S3-compatible storage; restore from the admin panel or command line	Must
AD-06	In-place upgrades with automatic database migrations and an automatic backup first	Must
AD-07	File storage on local disk or S3-compatible storage, with per-user quotas	Must
AD-08	SMTP settings with a test-send button	Must

ID	Requirement	Priority
AD-09	Command-line tool for recovery: reset owner password, run backup or restore, check health	Must
AD-10	phpBB importer: users, categories, topics, posts as closed messages, attachments, redirects from old URLs	Should
AD-11	Branding: logo, colours, custom CSS	Should
AD-12	Health page, structured logs and a metrics endpoint	Should

14. Security and privacy

All traffic is encrypted between client and server; content is readable by the server so it can be moderated and searched. There is no end-to-end encryption and no private messaging.

ID	Requirement	Priority
SP-01	HTTPS and secure WebSockets only (TLS 1.2 or later), certificates obtained and renewed automatically	Must
SP-02	Passwords hashed with Argon2id; sessions in secure, HTTP-only cookies; API tokens scoped and revocable	Must
SP-03	GDPR tooling: data export, erasure (messages anonymised or deleted, by user choice where law allows), consent records, retention settings	Must
SP-04	No third-party scripts, fonts or analytics by default	Must
SP-05	Content security policy, CSRF protection, rate limits on every endpoint, image metadata stripped on upload	Must
SP-06	Admin audit log of every settings, role and permission change	Must
SP-07	Published vulnerability disclosure policy and security-only patch releases	Must

ID	Requirement	Priority
SP-08	Media proxy: external images and link previews fetched by the server, hiding users' IP addresses	Should
SP-09	Retention policies: auto-delete messages older than N days, per category	Should

15. API, extensibility and federation readiness

The official web client uses the same public API as everyone else. Federation is not built in v1, but the data model and event system are shaped so it can be added without a rewrite.

ID	Requirement	Priority
AP-01	Public REST API documented with OpenAPI	Must
AP-02	Real-time event stream over WebSocket: message opened, appended, edited, closed; topic created, moved, archived	Must
AP-03	Personal and application tokens with scopes	Must
AP-04	Outgoing webhooks fire on closed messages and topic events; incoming webhooks post a closed message into a topic	Should
AP-05	Bot accounts that post through the API, shown with a bot badge	Should
AP-06	Plugin system with declared permissions, server hooks and interface slots	Could
FD-01	Every user, category, topic and message has a globally unique, stable URI	Must
FD-02	Internal events are recorded as activities (create, update, delete, announce) with actor, object and target, in an outbox	Must
FD-03	Content stored as Markdown source plus sanitised HTML	Must
FD-04	Each user and category gets a signing key pair at creation, for later server-to-server signatures	Should

ID	Requirement	Priority
FD-05	Data model carries a local/remote origin on users and content (all local in v1)	Should
FD-06	ActivityPub federation and chat bridges delivered as plugins	Could

16. Clients

The web app is the only client required for v1; native apps come later on the same API.

ID	Requirement	Priority
CL-01	Responsive web app for desktop and mobile browsers, installable as a progressive web app	Must
CL-02	Lightweight server-rendered read view for public pages, working without JavaScript	Must
CL-03	Keyboard shortcuts and a quick switcher for categories, topics and people	Should
CL-04	Offline reading of recently opened topics	Should
CL-05	Native iOS and Android apps that connect to any server by address	Could

17. Non-functional requirements

These targets are agreed; a load test before each release verifies them.

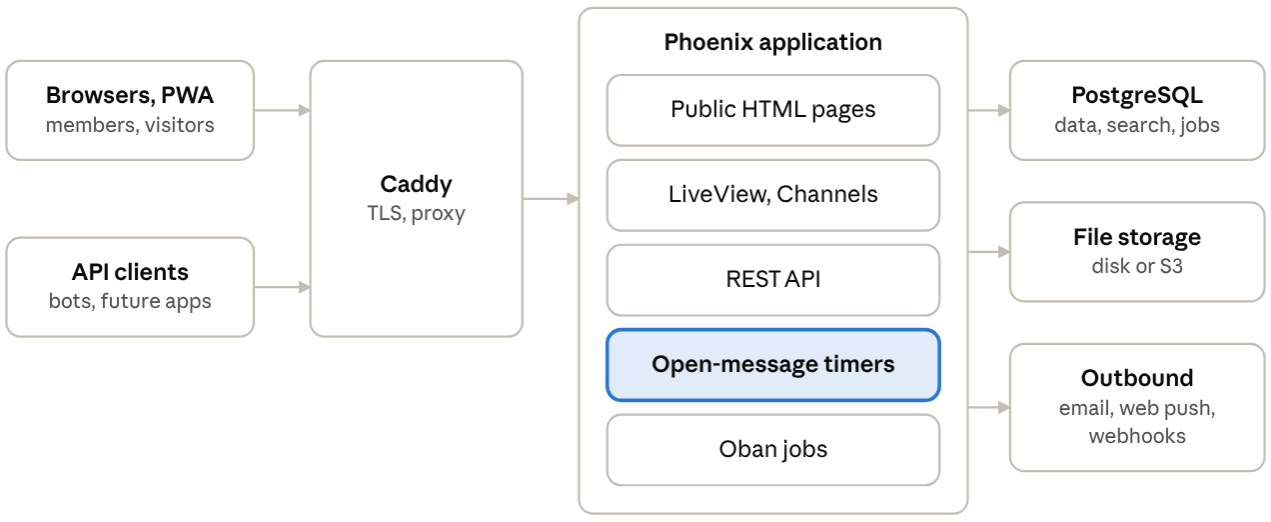
ID	Area	Target
NF-01	Live latency	New message or append shown to other viewers within 200 ms (95th percentile)
NF-02	Durability	A commit is stored in the database before it is acknowledged; a restart loses no sent text
NF-03	Page speed	Public topic page shows content in under 1 s on a mid-range phone over 4G
NF-04	Minimum server	2 vCPU, 2 GB RAM, 20 GB SSD for up to about 200 people online at once

ID	Area	Target
NF-05	Recommended server	4 vCPU, 8 GB RAM for up to about 1,000 people online at once
NF-06	Restart	Service back within 30 s; clients reconnect and catch up automatically
NF-07	Backups	Daily by default, 14 days kept
NF-08	Browsers	Current and previous version of major browsers, including mobile Safari
NF-09	Portability	Full export as JSON plus Markdown plus original attachments
NF-10	Quality	Automated tests cover every GM rule; accessibility audit and load test before each release
NF-11	Documentation	Admin, moderator, user and API guides shipped with each release

18. Proposed tech stack

Proposal: an Elixir/Phoenix application on PostgreSQL, behind Caddy, shipped as a Docker Compose bundle. It fits a server that holds thousands of live connections, runs a timer for every open message, and still serves plain HTML to readers and search engines — with only two required services besides the app.

One application serves pages, live updates and the API



architecture · Docker Compose bundle

Layer	Proposed choice	Why it fits
Language and framework	Elixir with Phoenix	Built for many long-lived WebSocket connections; Channels and Presence cover live delivery and “who is reading” (TP-04, TP-16); scales to several nodes with built-in PubSub
Growing messages	One lightweight process per open message, holding its window timer; every commit written to the database first	A timer per message costs almost nothing on the Erlang VM; after a restart, open messages are rebuilt from the database (NF-02)
Web interface	Server-rendered HTML for public pages; Phoenix LiveView for the signed-in app; small JavaScript hooks for the composer and keys	Readers need no JavaScript (AX-04, CL-02), pages are indexable (SE-04), one codebase instead of a separate front-end app
Database	PostgreSQL	Relational data and permission checks; built-in full-text and trigram search (SE-01)
Background jobs	Oban (job queue stored in PostgreSQL)	Emails, digests, auto-archive, backups, webhooks without adding Redis
Search	PostgreSQL full-text by default; adapter for Meilisearch or OpenSearch	No extra service for small servers (SE-07)
File storage	Local disk or S3-compatible object storage	AD-07
Reverse proxy and TLS	Caddy	Automatic certificates with minimal configuration (SP-01)
Email and push	SMTP via Swoosh; standard Web Push (VAPID)	Works with any mail provider; no vendor push relay (NO-05)
Authentication	Argon2id hashing, WebAuthn passkeys, TOTP, OpenID Connect client	SP-02, AC-03, AC-09

Layer	Proposed choice	Why it fits
Packaging	Docker Compose bundle (app, PostgreSQL, Caddy); standalone release for bare-metal installs	AD-01
Observability	Built-in telemetry with a Prometheus metrics endpoint	AD-12

Alternative considered: a Go back end with a TypeScript single-page app. It is equally capable, but needs a separate server-rendering path for readers and search engines, and more front-end tooling to maintain.

Building with Claude Code

The stack is confirmed on condition that Idejo can be built and maintained with Claude Code. Everything above runs from a terminal, which Claude Code needs. Elixir has less public code than JavaScript or Python, so written conventions and tests carry more weight in keeping generated code consistent.

ID	Requirement	Priority
DV-01	The repository holds a CLAUDE.md with architecture, conventions, commands and the requirement IDs, updated with every change	Must
DV-02	Every requirement ID has automated tests named after it (e.g. GM-03); all tests pass before a merge	Must
DV-03	Formatter, linter, type checks, tests and accessibility checks all run from the command line and in CI	Must
DV-04	Pinned versions of Elixir, Erlang, Phoenix and all dependencies; the full stack starts locally with one Docker Compose command	Must
DV-05	Small changes: one requirement or fix per pull request, reviewed by a person before merge (also supports LI-08)	Must
DV-06	Dependency audit on every build; human review of authentication, permission and upload code	Must

19. Licensing

Idejo is released under the GNU AGPL-3.0-only, with one additional term under section 7(b): an attribution notice that must be preserved. Mariano Pictet holds the copyright as steward until an organisation is formed, which then takes it over; contributions are taken under an agreement that keeps a later dual-licensing option open.

ID	Requirement
LI-01	Every source file carries an SPDX header (AGPL-3.0-only); the full licence text ships in the repository and the install
LI-02	Section 7(b) term: the attribution notice (product name and link to the project) shown in the interface's legal notices must be preserved in modified versions; wording in the NOTICE file: "Powered by Idejo — © Mariano Pictet and contributors — AGPL-3.0", linked to https://idejo.org
LI-03	AGPL section 13: every server shows a "Source code" link in its footer pointing to the exact source it runs, including local modifications
LI-04	Copyright steward: Mariano Pictet, as an individual, until an organisation is formed; the rights then transfer to it. A single steward keeps a second, commercial licence possible later
LI-05	Licence-grant CLA, individual and corporate, based on the Apache ICLA/CCLA templates and signed once through an automatic check on pull requests: contributors keep their copyright but grant the steward, and any successor organisation, a perpetual, worldwide, royalty-free copyright and patent licence, including the right to relicense. A sign-off (DCO) alone is not enough for dual licensing
LI-06	Only dependencies under licences compatible with both AGPL-3.0 and a future proprietary licence (e.g. MIT, BSD, Apache-2.0); checked automatically in CI

ID	Requirement
LI-07	Trademark policy for the name “Idejo” and its logo, held by the steward
LI-08	Human authorship of AI-assisted code: purely AI-generated code is not protected by copyright, while human selection, arrangement and modification are. So the steward writes the specifications (this document, kept in the repository), reviews and edits every change (DV-05), and AI-assisted commits carry a trailer such as “Assisted-by: Claude Code”. The AI tool’s terms must leave rights to its output with the user

The setup uses standard templates and common practice, so no custom legal drafting is needed now; a lawyer reviews it before any commercial licence is offered.

20. Items still to confirm

- [x] ~~All open questions are resolved; decisions are recorded in section 2.~~

Sources (licensing, section 19)

- CLA vs. DCO: which contributor agreement to use
- US copyright law and AI — Apache Software Foundation
- US Copyright Office reaffirms human authorship in AI works